A Illustrative example

The example below illustrates the approach to resolve a mention in a table to its corresponding entity, using an LLM. A high-level **instruction**, describing the entity linking task, is provided. A **question** contains the **mention** to be linked, together with a list of entity **candidates** extracted from a retriever, e.g., Wikidata Lookup Service or LamAPI. The candidates are provided in the following format: `<label [DESC] description [TYPES] type1, type2, ..., typeK>`. A snapshot of the **input** table is provided, i.e., N rows above and below the row containing the mention, in Markdown format. The LLM selects a candidate from the predefined list and outputs it verbatim as an **answer**.

Entity Linking Example

Instruction: *This is an entity linking task. The goal for this task is to link the selected entity mention in the table cells to the entity in the knowledge base. You will be given a list of referent entities, with each one composed of an entity name, its description and its type. Please choose the correct one from the referent entity candidates.* [...]

Input: `[TLE] List of high schools in South Dakota.`

```
col: |school|type|city|county|mascot|
row 0: |Aberdeen High School|Private|Aberdeen|Brown|Knights|
row 1: |Agar High School|Public|Agar|Sully|
[...]
row N: |Alcester-Hudson High School|Public|Alcester|Union|Cubs|
```

Question: *The selected entity mention is: 'Hyde'. The column is 'city'. The referent candidates are:* `<Dr. Jekyll and Mr. Hyde [DESC] fictional characters [TYPE] fictional characters group>` [...]
What is the correct referent entity for 'Hyde'?

Postilla: *Answer with just a candidate, selected from the provided referent entity candidates list, and nothing else. The selected candidate must be reported verbatim from the list provided as input. Each candidate in the list is enclosed between $<$ and $>$ and reports [DESC] and [TYPE] information.*

Answer: `<Hyde County [DESC] county in South Dakota, United States [TYPE] county of South Dakota>`

B Representative qualitative examples

We provide representative examples of the strengths and shortcomings of our method, reporting the input table, mention, candidate list, generated answers, and uncertainty estimates. Recoverable error 1 reports the standard case: a model's output is uncertain, with borderline accuracy (0.5), and our one-shot prediction highlights such uncertainty. Recoverable error 2 reports a more complex case, in which the same mention is referred over multiple seasons. As a consequence, the model struggles to fix on one single entity. Again, our method is able to recover this case. We also report an unrecoverable error case, in which there is no answer variability. Our method correctly predicts low answer variability, and the case is not marked for correction. However, the answer is wrong, and the error is not recovered. Notably, the regressor accurately captures uncertainty across all these examples.

Recoverable error 1: high answer variability → low accuracy

party	city	province	registration
Democracia Galega	Oleiros	A Coruña	1997-01-13
Partido Democrático Español	Madrid	Madrid	1997-01-13
Partido Nacional Republicano	Valladolid	Valladolid	1997-01-13
Partido del Amor Universal	Barcelona	Barcelona	1997-02-12
Els Verds de Formentera	Formentera	Balearic Islands	1997-02-20
Els Verds d'Eivissa	Ibiza	Balearic Islands	1997-02-20
	...		

Table: Registered political parties in Spain, mention highlighted

Name	Description	Type
Madrid	Spanish Congress Electoral District	electoral district of the Spanish Congress
Madrid	city in Iowa, United States	city in the United States
Madrid Province	province of Spain (1833-)	province of Spain
Madrid	constituency of the Senate of Spain	constituency of the Senate of Spain
Madrid	poem by Alfred de Musset	version, edition, or translation
Madrid	capital city of Spain	municipality of Spain
Madrid	None	passenger ship
Madrid	None	electoral district
Madrid	film	film
Madrid	mountain in South Africa	mountain
Madrid	encyclopedia article	encyclopedia article
	...	

Candidate entities with description and type (right and wrong answers)

Observed answers (count):

`<Madrid [DESC] capital city of Spain [TYPE] municipality of Spain>` **(5)**

`<Madrid [DESC] Spanish Congress Electoral District [TYPE] electoral district of the Spanish Congress>` **(3)**

`<Madrid [DESC] None [TYPE] electoral district>` **(2)**

Predictive Entropy:	0.649	Predicted using our method (avg):	0.588
Semantic Entropy:	0.478	Predicted using our method (avg):	0.421

Recoverable error 2: high answer variability → low accuracy

Season	Club	League	Apps	Goals
1986-87	Fiorentina	Serie A	5	1
1987-88	Fiorentina	Serie A	27	6
		...		
1994-95	Juventus	Serie A	17	8
1995-96	Milan	Serie A	28	7
		...		
1998-99	Inter	Serie A	23	6
1999-00	Inter	Serie A	18	6
		...		

Table: Roberto Baggio career stats, mention highlighted

Name	Description	Type
Serie A	top Italian football league	annual sporting event
Serie A (basketball) 2003-04	sports season	sports season
Serie A (basketball) 2007-08	None	sports season
Serie A	2nd tier of Italian women's rugby union	national championship
Serie A	top Italian pallapugno league	sports competition
1984-85 Serie A	sports season	sports season
1986-87 Serie A	sports season	sports season
1990-91 Serie A	sports season	sports season
1994-95 Serie A	sports season	sports season
1997-98 Serie A	sports season	sports season
1998-99 Serie A	sports season	sports season
1999-2000 Serie A	sports season	sports season
	...	

Candidate entities with description and type (right and wrong answers)

Observed answers (count):

<1986-87 Serie A [DESC] sports season [TYPE] sports season> **(4)**
<1994-95 Serie A [DESC] sports season [TYPE] sports season> **(3)**
<Serie A [DESC] top Italian football league [TYPE] annual sporting event> **(2)**
<1998-99 Serie A [DESC] sports season [TYPE] sports season> **(1)**

Predictive Entropy: 0.639 Predicted using our method (avg): 0.571
Semantic Entropy: 0.638 Predicted using our method (avg): 0.559

Unrecoverable error: no answer variability & zero accuracy

name	family	language	region	country	pop (M)
Agaw	Cushitic	Agaw	Horn of Africa	Ethiopia. Eritrea	1
Amhara	Semitic	Amharic	Horn of Africa	Ethiopia	20
Beja	Cushitic	Beja	Horn of Africa	Sudan. Eritrea	2
Bilen	Cushitic	Bilen	Horn of Africa	Eritrea	0.2
Gurage	Semitic	Gurage	Horn of Africa	Ethiopia	1.9
Oromo	Cushitic	Afan Oromo	Horn of Africa	Ethiopia. Somalia. Sudan. Kenya	30
Saho	Cushitic	Saho	Horn of Africa	Eritrea. Ethiopia	0.2
		...			

Table: Ethnic groups in Horn of Africa, mention highlighted

Name	Description	Type
Bilen i dag	Swedish periodical	periodical
Blågula Bilen	aid agency	aid agency
Bilen & miljön	Swedish periodical	periodical
Handsfreetelefoni i bilen	motion by Inger Jarl Beck et al. 2005	individual motion
Bilen people	ethnic group	ethnic group
Bilen	1992 film by John Goodwin	film
Ensam i bilen	article in Drömmen om bilen (1997)	article
Blin	language	modern language
När bilen drabbade landsbygden	article in Drömmen om bilen (1997)	article
Bilen	family name	family name
Mobilförbud i bilen	motion by Helena Bargholtz 2009	individual motion
	...	

Candidate entities with description and type (right and wrong answers)

Observed answers (count):

`<Bilen people [DESC] ethnic group [TYPE] ethnic group>` **(10)**

Predictive Entropy:	0.0	Predicted using our method (avg):	0.044
Semantic Entropy:	0.0	Predicted using our method (avg):	0.041

C Prompt segments distribution

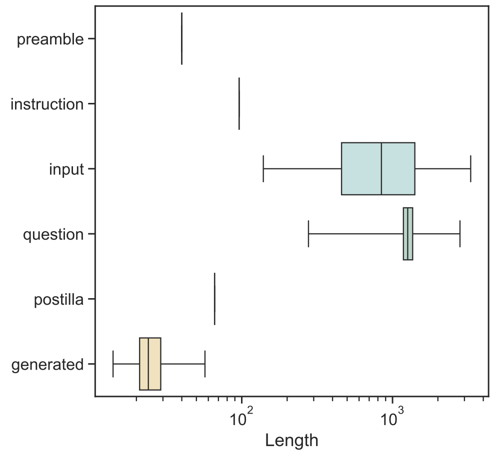

Fig. 9: Distribution of the number of tokens per prompt segment (TableLlama)

D Accuracy distribution

Figure 10 reports the accuracy distributions for the models utilized in the paper. The models are evaluated over $N = 10$ generation runs. Cases with zero accuracy can be further split by uncertainty according to Table 1.

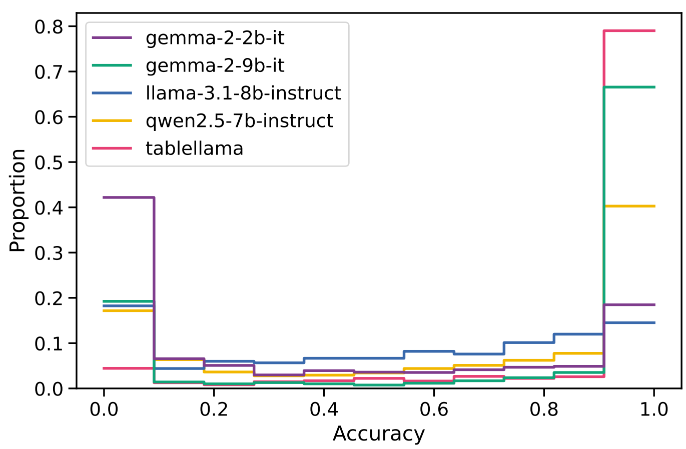

Fig. 10: Accuracy distribution over 10 non-deterministic generation runs derived from the same prompt

E Spearman correlation between estimated and true PE/SE

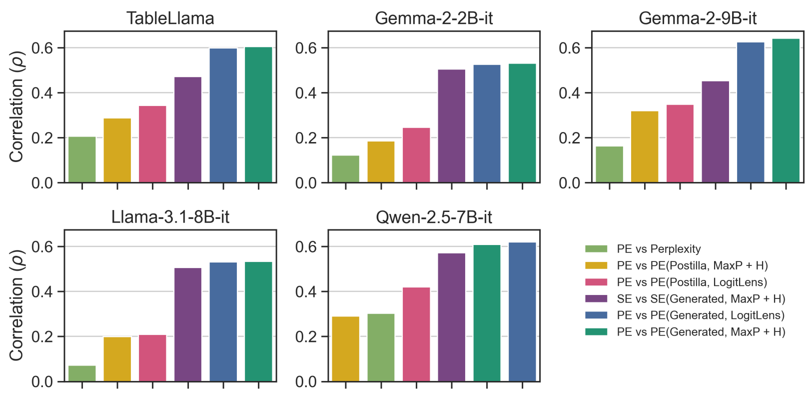

Fig. 11: Spearman's rank correlation coefficient (ρ) between the estimated entropy (PE or SE) and the multiple-generations entropy (PE or SE) across models. In the legend, the notation "`Target(Segment, Observable)`" indicates that the target variable *Target* was predicted using a regressor based on *Observable* features extracted from the *Segment* segment of the prompt.

F Answer variability as a function of temperature

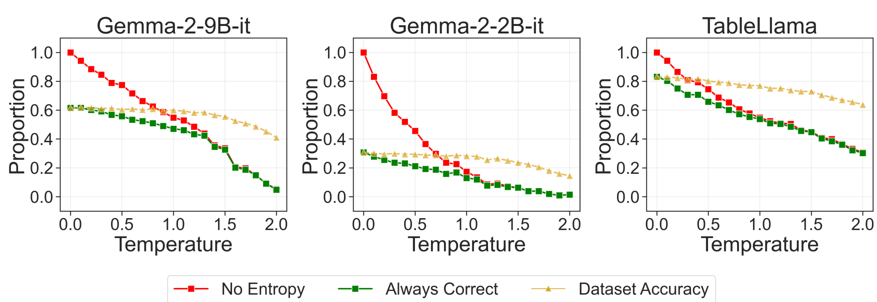

Fig. 12: Sensitivity to temperature of prompts with no output variation (*red*), prompts with correct answers only – still no output variation (*green*), and prompts with correct answer on average (≥ 0.5, *yellow*), **(left)** Gemma-2-9b-it, **(center)** Gemma-2-2b-it, and **(right)** TableLlama.

To evaluate the variability of answers and recoverable cases, we systematically sweep temperature values in the range $0.0 \leq \tau \leq 2.0$ with steps of 0.1 to assess the temperature effect on (1) the number of items that show output variability and (2) the average accuracy. For capacity reasons, we utilize a subset of 200 elements from the original dataset. Figure 12 shows the results of the experiment for two selected LLMs. The *yellow* series shows average accuracy, highlighting how higher τ values deteriorate the performance on the task. At the same time, lower τ settings yield the highest proportion of always correct cases (*green* line). However, again at low τ, output variability is also lowest; the uncertainty of these cases cannot be estimated, making this setting unsuitable for recovery. Operationally, a reasonable trade-off for τ is to minimize the accuracy loss while maximizing the number of cases recoverable through uncertainty, that is, those above the *red* line. Another way to view this trade-off is by examining the difference between the *red* and *green* lines: this difference represents the proportion of unrecoverable cases, that is, cases that are always wrong and have no output variability. Making these cases recoverable requires "paying" by reducing always correct cases, while trying to preserve average accuracy (*yellow*) as much as possible. This experiment illustrates how temperature can be adjusted depending on specific combinations of model, task, and application constraints.

G Transformer-based time complexity

To further assess the practicality and usability of our approach, in this section we derive the time complexity of a Transformer-based architecture, such as the ones used by the models considered in this work. We consider a Transformer with L layers, and d hidden size. The context length is set to N, while the number of generated tokens is G. The time complexity of a single forward pass over a prompt $X \in \mathbb{R}^{N \times d}$can be decomposed into the following components:

- **Self-attention:** Given the $Q, K, V \in \mathbb{R}^{N \times d}$ matrices, the time complexity of the self-attention mechanism is $O(N^2 \cdot d)$, where N is the context length and d is the hidden size. The attention scores are computed as QK^T, and the output is computed as $\sigma(QK^T)V$. The time complexity of this operation is $O(N^2 \cdot d)$.
- **Feed-forward:** During the feed-forward step, the time complexity is $O(8N \cdot d^2) = O(N \cdot d^2)$ overall, where d is the hidden size. In this we can include also the projection of the input to the QKV space and the final projection to the output space.

In total one has a time complexity of $O(N^2 \cdot d + N \cdot d^2)$ for a single Transformer layer, which becomes $O(L[N^2 \cdot d + N \cdot d^2])$, where L is the number of layers.

If we now suppose to generate G tokens without the use of a KV-cache, the time complexity of the g-th generation step is $O(L[(N + g)^2 \cdot d + (N + g) \cdot d^2])$, for every $g \in [G]$. If we then sum over all the generations, we have:

$$O\left(\sum_{g=1}^{G} L\left[(N+g)^2 d + (N+g)d^2\right]\right) = \tag{4}$$

$$O\left(L\sum_{g=1}^{G} \left[(N+g)^2 d + (N+g)d^2\right]\right) = \tag{5}$$

$$O\left(L\sum_{g=1}^{G} \left[(N^2 + 2Ng + g^2)d + (N+g)d^2\right]\right) = \tag{6}$$

$$O\left(L\left[(GN^2 + 2NG^2 + G^3)d + (GN + G^2)d^2\right]\right) = \tag{7}$$

$$O\left(L\left[G(N+G)^2 d + G(N+G)d^2\right]\right) = \tag{8}$$

$$O\left(LG\left[(N+G)^2 d + (N+G)d^2\right]\right) \tag{9}$$

which shows that the time complexity of generating G tokens is quadratic in the number of overall tokens $N+G$, when $G \ll N$, otherwise it would become cubic in the number of generated ones.

When a KV-cache is used, while the time for processing the prompt is the same, a major computational saving is obtained during the generation phase. In this case, the time complexity of the g-th generation step can be decomposed into the following components:

- **Self-attention:** During the self-attention, Q reduces to a single vector $q_g \in \mathbb{R}^{1\times d}$, while K, V becomes $K_g, V_g \in \mathbb{R}^{(N+g-1)\times d}$. Overall, the time complexity of this operation is $O((N+g)\cdot d)$.
- **Feed-forward:** Since the output of the Self-Attention has a dimension of $1 \times d$, the feed-forward step has a time complexity of this operation is $O(d^2)$, which includes the projection of the input to the QKV space and the final projection to the output space.

If we then sum over all the generations $g \in [G]$, we have:

$$O\left(\sum_{g=1}^{G} L\left[(N+g)d + d^2\right]\right) = \tag{10}$$

$$O\left(L\sum_{g=1}^{G} \left[(N+g)d + d^2\right]\right) = \tag{11}$$

$$O\left(L\left[G(N+G)d + Gd^2\right]\right) = \tag{12}$$

$$O\left(LG\left[(N+G)d + d^2\right]\right) \tag{13}$$

which shows that the time complexity of generating G tokens is linear in the number of overall tokens $N+G$, when $G \ll N$, otherwise it would become quadratic in the number of generated ones.

Table 2: Time complexity of the Transformer-based architecture. L is the number of layers, N is the number of tokens in the prompt, G is the number of generated tokens, and d is the hidden size.

Phase	**Without KV-cache**	**With KV-cache**
Prompt processing	$O(L[N^2 \cdot d + N \cdot d^2])$	$O(L[N^2 \cdot d + N \cdot d^2])$
Generation	$O(LG[(N+G)^2 d + (N+G)d^2])$	$O(LG[(N+G)d + d^2])$

This simple derivation shows that our proposed approach to learn to estimate the multiple-generations Predictive Entropy (PE) is still computationally promising even when advanced KV-cache techniques are employed during inference.